CampusGpt

Final Year Project

Session 2022-2025

A project submitted in partial fulfillment of the degree of

BS in Data Science

Department of Software Engineering

Faculty of Computer Science & Information Technology

The Superior University, Lahore

Fall 2025

Type (Nature of project)	[ ] **D**evelopment [ ] **R**esearch [ ] **R&D**
Area of specialization	
FYP ID	FYP-BSDSM-F25-005

Project Group Members

Sr.#	Reg. #	Student Name	Email ID	*Signature
(i)	Su92-bsdsm-s24001	Muhammad Babar	Su92-bsdsm-s24-001@superior.edu.pk	
(ii)	Bdsm-s22-005	Ahmad Mamoon	Bdsm-S22-005@superior.edu.pk	
(iii)	Bdsm-s22-001	Fraz Shayna	Bdsm-s22-001@superior.eud.pk	

*The candidates confirm that the work submitted is their own and appropriate credit has been given where reference has been made to work of others

Plagiarism Free Certificate

This is to certify that, I **Muhammad Babar** S/D of **Aass Muhammad**, group leader of FYP under registration no __ Su92-bsdsm-s24-001___at Software Engineering Department, The Superior University, Lahore. I declare that my FYP report is checked by my supervisor.

Date: Name of Group Leader: _____ Signature: _____

Name of Supervisor: Dr.H.M.Shahzad Dar Co-Supervisor: Dr. H. M. Shahzad Dar

Designation: Assistant Professor Designation: Assistant Professor

Signature: _____ Signature: _____

HoD: Dr. Tehreem Masood

Signature: _____

APPROVAL

PROJECT SUPERVISOR

Comments: _____

_____

Name:_____

Date:_____ Signature:_____

PROJECT MANAGER

Comments: _____

_____

Date:_____ Signature:_____

HEAD OF THE DEPARTMENT

Comments: _____

_____

Date:_____ Signature:_____

Dedication

This project is dedicated to my teachers, mentors, and academic guides, whose continuous support, encouragement, and wisdom have played a significant role in shaping my learning journey. Their dedication to teaching, their belief in my abilities, and their constant motivation have inspired me to work hard, think creatively, and pursue excellence throughout this project.

I also dedicate this work to my classmates, peers, and friends who stood beside me with support, discussion, and positivity during the development of this system. Their collaboration and encouragement helped me stay focused and overcome challenges at every step.

Finally, this project is dedicated to all future learners, innovators, and technology enthusiasts who aim to build solutions that create real impact. May this work serve as an inspiration for them to explore new ideas, embrace innovation, and contribute meaningfully to the advancement of technology and education.

Acknowledgements

First and foremost, I would like to express my sincere gratitude to my project supervisor, **Dr. Muhammad Shahzad Dar**, for her continuous guidance, support, and encouragement throughout the course of this project. Her valuable advice, constructive feedback, and patience have helped me overcome challenges and complete this work successfully.

I am also thankful to my co-supervisor, **Dr. Hafiz Muhammad Shahzad**, for his insightful suggestions and guidance that enhanced the quality of my research and development process. Their expertise and dedication have been truly inspiring.

I would like to extend my appreciation to the faculty and staff of the Software Engineering Department at The Superior University, Lahore, for providing a supportive and motivating environment for learning and research.

My heartfelt thanks go to my classmates and friends for their cooperation, support, and encouragement during the project. Their ideas, discussions, and motivation have played a key role in keeping me focused and driven.

Finally, I would like to thank all those who directly or indirectly helped me in completing this project. Your support, guidance, and encouragement have been invaluable, and I am deeply grateful for your contributions.

This project would not have been possible without the assistance and inspiration of all these people, and I dedicate my sincare thanks to each one of them

Table of Contents

CampusGpt i
Plagiarism Free Certificate ii
Dedication iv
List of Figures x
List of Tables xi
Chapter 1 1
Introduction 1
1.1 Background and Motivation 2
1.2 Problem Statement 3
1.3 Objectives of the Study 3
1.4 Research Questions 3
1.5 Scope and Limitations ..... **Error! Bookmark not defined.**
1.6 Thesis Organization 4
Chapter 2 6
Literature Review 6
Chapter 2: Literature Review 7
2.1 Introduction 7
2.2 Overview of Campus Management Systems 7
2.3 AI in Education and ERP Systems 8
2.4 Natural Language Processing for Campus Assistance 9
2.5 Chatbots and Virtual Assistants in Academic Settings 9
2.6 Predictive Analytics in Education 10
References 13
Chapter 3 16
Research & Methodology 16
3.1 System Architecture 17
3.1.1 User Interface Layer 17
3.1.2 Application Layer 17
3.1.3 AI and Automation Layer 17
3.1.4 Database Layer 18
3.1.5 Integration and API Layer 18
3.2 Functional Modules of CampusGPT 18
3.2.1 Student Module 18
3.2.2 Teacher Module 19

3.2.3 Admin Module 20
3.2.4 Staff Module 21
3.3 AI Components and Agentic Workflows 22
3.3.1 NLP-Based Academic Advisor 22
3.3.2 Attendance Analyzer & Risk Prediction 23
3.3.3 CGPA/SGPA Forecasting Engine 23
3.3.4 Automated Quiz & Assignment Generator 24
3.4 System Modules and Features 24
3.4.1 Student Module ..... **Error! Bookmark not defined.**
3.4.2 Teacher Module 26
3.4.3 Admin Module 26
3.4.4 Staff Module 27
3.5 Development Tools and Technologies 27
3.5.1 Front-End Technologies 27
3.5.2 Back-End Technologies 28
3.5.3 Database Technologies 28
3.5.4 MySQL / PostgreSQL (Optional) 28
3.5.5 Development & Deployment Tools 29
3.5.6 Security Technologies 29
3.6 Experimental Setup and Testing 30
3.6.1 Experimental Environment 30
3.6.2 Dataset Setup 30
3.6.3 Testing Procedures 31
3.6.4 User Testing 32
3.6.5 Performance Testing 32
3.6.6 Bug Fixing & Iterative Improvement 32
3.7 Evaluation Metrics 32
3.7.1 AI Model Evaluation Metrics 32
3.7.2 NLP Response Evaluation 33
3.7.3 System Performance Metrics 34
3.7.4 Usability Metrics 34
3.7.5 Reliability Metrics 34
Chapter 4 36
Experimental Results 36
4.1 System Implementation Results 36

4.2 Functional Testing and Validation 37
4.2.1. Student Module Testing 37
4.2.2 Teacher Module Testing 37
4.2.3 Admin Module Testing 37
4.2.4 Staff Module Testing 38
4.2.5 AI Components Validation 38
4.2.6 Overall System Validation 38
4.3 Performance Analysis 39
4.3.1 Load Handling and Scalability 40
4.3.3 AI Model Performance 41
4.3.4 User Experience and Interface Performance 42
4.3.4 System Reliability 42
4.4 AI Model Accuracy and User Interaction Analysis 43
4.4.1 NLP Academic Advisor Accuracy 43
4.4.2 CGPA/SGPA Prediction Model Accuracy 43
4.4.3 Attendance Risk Analyzer Accuracy 44
4.4.4 Quiz and Assignment Generator Quality 45
4.4.5 User Interaction Analysis 45
4.4.6 System Interaction Metrics 46
4.5 Discussion of Results 48
5.1 Summary of Findings 51
5.2 Contributions of the Study 51
5.2.1 Development of an AI-powered Campus ERP System 52
5.2.2 Introduction of an NLP-Based Academic Advisor 52
5.2.3 Automated Quiz & Assignment Generator 52
5.2.4 Predictive Analytics for Academic Performance 52
5.2.5 Improved Campus Communication and Automation 52
5.2.6 Modern UI/UX for All User Roles 52
5.2.7 Scalable Architecture for Future Expansion 52
5.3 Limitations 53
5.3.1 Limited Real University Dataset 53
5.3.2 Dependence on Internet Connectivity 53
5.3.3 Limited Biometric or Hardware Integration 53
5.3.4 NLP Advisor Accuracy Depends on Dataset 53
5.3.5 Performance Under Very High Load 53

5.3.6 Limited Mobile App Support 53
5.3.7 Manual Data Entry Required Initially 54
5.4 Suggestions for Future Enhancements 54
5.4.1 Integration of Biometric and IoT Devices 54
5.4.2 Development of a Mobile Application 54
5.4.3 Expanded AI Training with Real University Data 54
5.4.4 Voice-Based Virtual Assistant 54
5.4.5 Advanced Analytics and Reporting Tools 55
5.4.6 Integration with Learning Management Systems (LMS) 55
5.4.7 Offline Mode for Low-Internet Areas 55
5.4.8 Multi-Language Support 55
5.4.9 Chat-Based Workflow Automation 55
5.4.10 Cloud Auto-Scaling 55

List of Figures

Figure 3.1 : CampusGpt System Overview Architecture....17
Figure 3. 2 : CampusGpt Student Module Workflow 19
Figure 3.3: CampusGPT Teacher Module Workflow 20
Figure 3.4: Admin Module Management Overview 21
Figure 3. 5: Staff Module Workflow Diagram....22
Figure 3. 6: NLP Academic Advisor Workflow....23
Figure 3.7: Attendance Risk Prediction Model 23
Figure 3. 8: CGPA Prediction Process Flow....24
Figure 3. 9: Automated Quiz Generation Pipeline 24
Figure 3. 10: Student Dashboard Interface....26

Figure 4. 1: Performance Analysis 40
Figure 4. 2: Load Handling and Scalability 41
Figure 4. 3: AI Model Accuracy and User Interaction Analysis 44
Figure 4. 4: Attendance Risk Analyzer Accuracy....45
Figure 4. 5: Model Routing Distribution (Gemini vs Gork)....47
Figure 4. 6: Tools Accuracy 47
Figure 4. 7: AI Agent Response Time Comparison (Gemini vs Groq)....49

List of Tables

Table 3.1 Description of Functional Modules in CampusGpt 19
Table 3.2 Student Portal Features and Their Functionalities..... 20
Table 3.3 Teacher Portal Features and Their Functionalities 21
Table 3.4 Admin Portal Features and Management Controls..... 22
Table 3.5 Staff Portal Features Overview..... 23
Table 3.6 AI Components Used in CampusGpt and Their Purpose..... 24
Table 3.7 Dataset and Knowledge Base Sources 25
Table 3.8 Tools, Frameworks, and Technologies Used in Development..... 26
Table 3.9 Security & Role-Based Access Control Matrix 27
Table 3.10 Testing Methods and Evaluation Parameters 28

Chapter 1

Introduction

Chapter 1: Introduction

1.1 Background and Motivation

The digital transformation of educational institutions has become an essential requirement in modern times. Universities today manage thousands of students, teachers, staff members, and administrative tasks on a daily basis.[1] These activities include attendance management, course handling, academic records, student queries, timetable updates, fee management, and communication between different departments. However, many universities still rely on outdated or semi-digital systems that are difficult to use, lack automation, and provide limited support to users. [20] As a result, students, teachers, and administrators often face delays, confusion, and inefficiencies in performing even basic academic operations.

With the rapid advancement in Artificial Intelligence (AI), Natural Language Processing (NLP), and automation technologies, there is a significant opportunity to redesign university systems with smarter, more interactive solutions. Modern students expect quick access to academic information, personalized support, deadline reminders, and clear academic insights. Teachers also require tools that can reduce their workload, such as automated quiz generation, quick attendance marking, and simplified communication with students. Similarly, administrative departments need systems that provide accurate analytics, automated reporting, and a centralized platform to manage all campus-related operations.

CampusGPT is motivated by the need to solve these challenges and bring intelligence into the heart of the university management system. It introduces an AI-powered, agentic ERP system that not only handles routine tasks but also performs intelligent actions such as predicting student performance, analyzing attendance risks, generating quizzes from lecture materials, summarizing notes, and assisting students through a smart academic advisor. Instead of searching through complex menus or waiting for manual responses, users can simply interact with the system through a conversational AI interface. [6]

The motivation behind developing CampusGPT also arises from the common issues faced by students, such as missing deadlines, difficulty in understanding course content, limited access to academic guidance, and challenges in tracking their academic progress. Teachers face time-consuming manual processes, while administrators deal with fragmented data and slow workflows. CampusGPT aims to unify all these operations under one intelligent platform to enhance productivity, efficiency, and overall campus experience.

In summary, this project is driven by the vision of transforming traditional university systems into intelligent, automated, and student-centered environments. CampusGPT seeks to serve as a modern solution that integrates AI, automation, and ERP capabilities to support students, teachers, staff, and administrators in a seamless and efficient manner. [4]

1.2 Problem Statement

Most universities still use old, outdated ERP systems that do not support automation, intelligence, or real-time assistance. Students struggle to access information such as timetables, attendance, course details, deadlines, and academic progress. Teachers face extra workload due to manual quiz creation, grading, attendance management, and communication with students. Administrators also deal with slow processes, scattered data, and limited reporting tools. There is no smart system that can guide students, support teachers, or help admin staff through AI. Therefore, there is a strong need for an intelligent, unified, and automated ERP system that integrates AI capabilities to support academic, administrative, and learning activities in universities.

1.3 Objectives of the Study

The main objective of this study is to develop an AI-based, agentic ERP system that improves campus management and provides intelligent support to students, teachers, and administrators. This project aims to create a smart platform where students can easily access academic information, receive reminders, and get AI-generated assistance. Another objective is to help teachers by automating tasks such as quiz generation, attendance management, and student performance analysis. The system also aims to support administrators through centralized data, automated reports, and predictive analytics. Overall, the goal is to build a modern, user-friendly, and fully automated campus system that increases efficiency and enhances the overall academic experience.

1.4 Research Questions

This study is guided by the following research questions:

1. How can an AI-powered, agentic ERP system improve the efficiency of campus management compared to traditional university systems?
2. Can an AI academic advisor help students by providing instant academic support, personalized guidance, and accurate predictions of their academic progress?

3. How effectively can AI automate teacher tasks such as quiz generation, attendance management, and performance analysis?
4. To what extent can predictive analytics (CGPA prediction, attendance risk detection, student performance alerts) improve decision-making for students, teachers, and administrators?
5. How can a unified AI-based system enhance communication and reduce workload across students, teachers, staff, and administration?

1.5 Scope of the Study

This project focuses on the design and development of CampusGPT, an AI-powered and agentic ERP system intended for use within a university environment. The system is designed to support students, teachers, administrative staff, and management by providing intelligent automation and centralized campus services.

The scope of this study includes the development of an AI-based academic advisor capable of answering campus-related and academic queries using natural language processing. The system provides separate portals for students, teachers, administrators, and staff, each offering role-specific functionalities such as attendance management, course handling, announcements, academic records, and communication tools.

The project also covers the integration of artificial intelligence features such as automated quiz and assignment generation, lecture note summarization, CGPA and SGPA prediction, and attendance risk analysis. Interactive dashboards are developed to display academic progress, notifications, deadlines, and analytics in a clear and user-friendly manner. In addition, an automation layer is implemented to generate reminders, alerts, and reports to support academic decision-making.

Security and access control are also part of the scope, ensuring secure authentication, role-based authorization, and reliable cloud-based data storage. Overall, the system aims to improve campus communication, reduce manual workload, and provide intelligent, automated support across university operations.

1.6 Limitations of the Study

Despite offering advanced features and intelligent automation, CampusGPT has certain limitations. The accuracy of AI-driven features depends heavily on the quality, availability, and completeness of university-provided data. Incomplete or outdated data may affect prediction accuracy and response quality.

Real-time integration with existing university systems may require additional APIs, infrastructure compatibility, and administrative permissions, which are outside the scope of this study. Predictive models

such as CGPA forecasting and attendance risk analysis may not always achieve perfect accuracy due to variations in teaching methods, assessment criteria, and institutional policies.

Some AI functionalities, including advanced natural language processing and automated content generation, require higher computational resources, which may impact performance on low-resource environments. Additionally, the system does not cover advanced financial auditing or complex human resource management operations beyond basic staff management.

1.7 Thesis Organization

This thesis is organized into five chapters, each explaining a specific part of the research and development of the CampusGPT system.

Chapter 1 – Introduction

This chapter provides an overview of the project, including the background and motivation, problem statement, objectives, research questions, scope, and limitations of the study.

Chapter 2 – Literature Review

This chapter discusses previous research related to campus management systems, educational ERPs, AI in education, natural language processing, Chabot's, and automation agents. It also identifies existing gaps that motivate the need for an AI-driven campus system.

Chapter 3 – Methodology

This chapter explains the complete design and development process of CampusGPT. It covers the system architecture, functional modules, AI workflows, dataset and knowledge base, algorithms, use case diagrams, DFDs, ERD, technologies used, and system security.

Chapter 4 – Results and Discussion

This chapter presents the implementation results, system screenshots, AI performance evaluation, prediction accuracy, automation testing, user interaction analysis, and overall discussion of the findings.

Chapter 5 – Conclusion and Future Work

This chapter concludes the study by summarizing the main outcomes, contributions, and limitations. It also suggests future improvements and possible extensions of the CampusGPT system.

Chapter 2
Literature Review

Chapter 2: Literature Review

2.1 Introduction

A literature review provides an overview of previous studies, technologies, and research related to the topic. It helps understand how different systems work, what challenges exist in current solutions, and how modern technologies such as AI and automation are being used in educational environments.

In the context of universities, campus management systems have been used for many years to manage student records, attendance, fee details, and administrative tasks. However, most traditional ERPs lack intelligence, automation, and personalization. With the rise of Artificial Intelligence (AI), Natural Language Processing (NLP), and data-driven decision-making, researchers have explored new ways to enhance student support, academic guidance, campus operations, and communication systems.

This chapter reviews existing work in areas such as campus management systems, AI in education, virtual assistants, chatbots, predictive analytics, and automation agents. It highlights their strengths, limitations, and the technological gap that motivates the development of CampusGPT—an AI-powered, agentic ERP designed to modernize and automate university operations.

2.2 Overview of Campus Management Systems

Campus Management Systems (CMS), also known as University ERP systems, are software platforms designed to manage academic and administrative activities within an educational institution. These systems typically include modules for student records, attendance, course registration, fee management, class schedules, and examination handling. Their main purpose is to centralize data and simplify daily campus operations.

Traditional CMS tools are mostly rule-based and rely on manual data entry. They provide basic functionality such as storing student information, generating reports, and displaying course-related details. While they help automate some tasks, they often lack flexibility, user-friendliness, and advanced decision-making support. Many existing systems do not offer real-time notifications, personalized services, or intelligent features that modern students and teachers now expect.

Another limitation of conventional systems is the absence of integrated communication tools. Students usually need separate platforms for announcements, academic queries, and course materials. Teachers also struggle with manual tasks such as preparing quizzes, marking assignments, and managing student performance data. Administrative staff face challenges when handling large volumes of information and generating timely reports.

Recent developments in education technology have highlighted the need for smarter, AI-enabled ERP systems. These advanced solutions aim to improve user experience, automate repetitive tasks, provide personalized insights, and support data-driven decision-making. CampusGPT builds on this concept by integrating AI agents, automation workflows, and predictive analytics into a unified campus platform, offering a modern alternative to traditional CMS systems.

Popular ERP systems used in higher education include Oracle PeopleSoft Campus Solutions and Oracle Campus Solutions. These systems provide modules for student information systems (SIS), human resources, finance, and course management. Studies show that ERP systems help improve data consistency and reduce paperwork; however, they are often complex, expensive, and difficult to customize for specific institutional needs [1].

In Pakistan, many universities use locally developed systems or partial ERP solutions that mainly focus on attendance, fee management, and result generation. These systems usually lack advanced analytics, automation, and intelligent support for students and teachers.

2.3 AI in Education and ERP Systems

Artificial Intelligence (AI) has become one of the most influential technologies in modern education. It is being used to improve learning experiences, automate administrative tasks, and support decision-making within academic institutions. AI allows educational systems to move beyond simple data storage and deliver personalized, intelligent, and interactive services to students, teachers, and administrators.

In recent years, universities have started integrating AI into their ERP systems to enhance efficiency and reduce manual workload. AI-powered systems can analyze large amounts of educational data, identify patterns, predict outcomes, and provide real-time insights [2]. For example, AI can help predict student performance, detect attendance risks, recommend learning resources, and automate communication between teachers and students.

AI also supports personalized learning. Through Natural Language Processing (NLP) and machine learning algorithms, systems can answer student queries, generate customized study plans, and offer explanations based on a student's academic history. This level of personalization is not possible with traditional ERP systems.

In administrative areas, AI contributes by automating routine processes such as fee reminders, report generation, approval workflows, and timetable management. It enhances accuracy and speeds up decision-making, helping universities operate more efficiently.

The integration of AI within ERP systems marks a shift from static, menu-based platforms to dynamic, intelligent environments. This sets the foundation for advanced systems like CampusGPT, which combines AI agents, automation, and predictive analytics to deliver next-generation campus management.

2.4 Natural Language Processing for Campus Assistance

Natural Language Processing (NLP) is a key field of Artificial Intelligence that enables computers to understand, interpret, and respond to human language. In educational environments, NLP plays an increasingly important role by making campus systems more interactive, user-friendly, and accessible. Instead of navigating complex menus or searching through multiple portals, users can simply type or speak their query and receive an instant response [3] [21].

NLP-based systems can assist students by answering questions related to timetables, attendance, course outlines, assignment deadlines, and exam preparation. These systems can also summarize lectures, generate study notes, and provide explanations for academic topics. This reduces dependency on teachers for routine questions and gives students immediate support whenever needed.

For teachers, NLP can simplify tasks such as generating quizzes, drafting announcements, and communicating with students. It can analyze lecture materials and automatically create relevant questions or summaries. This not only saves time but also improves consistency and quality of academic content.

NLP also benefits university administration by automating communication workflows, answering frequently asked questions, and providing quick access to records and reports. It reduces the number of repetitive inquiries and allows staff to focus on more important tasks.

In modern campuses, NLP-enabled chatbots and virtual assistants are becoming essential tools. They offer 24/7 availability, personalized responses, and efficient problem-solving. CampusGPT uses NLP as a core component to interact with students and staff, making it a smart, conversational, and highly accessible ERP system.

2.5 Chatbots and Virtual Assistants in Academic Settings

Chatbots and virtual assistants have become increasingly common in modern educational institutions. They provide an efficient way to support students, teachers, and staff by answering questions, automating tasks, and offering instant access to important information. Unlike traditional systems that require manual navigation, virtual assistants respond directly to user queries through natural conversation, making them more convenient and user-friendly.

In academic settings, chatbots help students with tasks such as checking timetables, viewing attendance, understanding course content, tracking deadlines, and receiving reminders. They reduce the workload on teachers and administrative staff by handling repetitive queries, allowing humans to focus on more meaningful academic and administrative activities. [17] [22]

For teachers, virtual assistants can help generate quizzes, share class updates, analyze student performance, and assist in classroom management. They also enable faster communication with students, ensuring that important notifications reach everyone on time.

Administrative departments use chatbots to automate common requests, such as fee inquiries, event announcements, and campus updates. This improves communication and reduces delays in information delivery.

Modern AI-based assistants go beyond simple question-answer capabilities [4] [15] . They can learn from user behavior, provide personalized recommendations, predict academic outcomes, and guide students based on their progress. Systems like CampusGPT extend these capabilities by integrating chat-based interactions with campus ERP functionalities, creating a unified and intelligent support system for the entire university.

2.6 Predictive Analytics in Education

Several studies and software solutions have explored the use of ERP systems, AI tools, and chatbots in educational environments. Traditional university ERPs such as PeopleSoft, Oracle Campus Solutions, and local systems used in Pakistani universities typically focus on basic tasks like attendance, fee management, course registration, and academic record handling. While these systems help centralize data, they are often complex, rigid, and lack intelligent automation.

Recent research has introduced AI-based features such as chatbots for answering student queries, machine learning models for performance prediction, and automated grading tools. Some institutions have experimented with virtual assistants for academic guidance, while others have developed dashboards for analytics. However, most existing solutions are limited in scope—either they support only one department, offer only basic chatbot functionality, or lack integration with the full ERP system.

Predictive analytics uses machine learning techniques to analyze historical data and forecast future outcomes. In education, predictive models are widely used to predict student performance, dropout risk, and academic success.

Several studies have applied machine learning algorithms such as decision trees, logistic regression, and neural networks to predict student grades and identify at-risk students [5] [6] . These models help educators

intervene early and improve learning outcomes. However, most predictive systems are implemented as standalone tools and are not integrated with real-time ERP data.

A major gap in current research and systems is the absence of a unified, agentic, AI-powered campus platform that combines:

2.6.1 Full ERP Functionalities

Existing university ERP systems mainly focus on basic administrative operations such as attendance management, fee records, course registration, and examination results. While these systems help in centralizing institutional data, they often operate as static platforms with limited flexibility. Most traditional ERPs lack intelligent decision-making capabilities, real-time analytics, and automation support. Additionally, they are usually complex, expensive, and difficult to customize according to institutional needs. This limitation highlights the need for a modern ERP system that not only manages academic and administrative data but also supports intelligent processing and automation within a unified environment.

2.6.2 Intelligent Academic Advisor

An intelligent academic advisor is one of the most critical missing components in traditional campus management systems. Current systems do not provide interactive academic guidance or personalized support for students. Research shows that AI-powered advisors can assist students by answering academic queries, guiding course selection, explaining policies, and providing instant academic support. However, most existing solutions implement academic advising as standalone chatbots without access to real-time student data. CampusGPT bridges this gap by integrating an AI-based academic advisor directly with ERP data, allowing it to provide personalized, accurate, and context-aware guidance to students and teachers.

2.6.3 Predictive Analytics in Education

Predictive analytics plays a vital role in modern educational systems by analyzing historical academic data to forecast future outcomes. Several studies have applied machine learning techniques to predict student performance, identify dropout risks, and analyze academic trends. These models help institutions take early corrective actions and improve student success rates. Despite their effectiveness, most predictive analytics tools are developed as independent systems and are not integrated with live ERP data. CampusGPT addresses this limitation by embedding predictive models such as CGPA prediction and attendance risk analysis directly into the campus ERP, enabling real-time academic insights and proactive decision-making.

2.6.4 Automation Agents in Campus Systems

Automation agents are designed to perform routine tasks without human intervention, reducing workload and operational delays. In educational environments, automation can be used for sending reminders, generating reports, managing schedules, and triggering alerts based on predefined conditions. Existing

campus systems either lack automation or provide very limited workflow automation. Research on smart campuses highlights automation mainly in infrastructure and IoT applications rather than academic intelligence. CampusGPT introduces AI-driven automation agents that operate across academic and administrative domains, enabling efficient, timely, and intelligent task execution throughout the campus.

2.6.5 Personalized Recommendations

Personalization is a key expectation of modern digital systems, yet it is largely absent in traditional educational ERPs. Most systems provide the same information and interfaces to all users, regardless of their academic background or needs. AI-based recommendation systems can analyze student behavior, academic performance, and engagement patterns to suggest relevant learning resources, courses, and academic actions. CampusGPT incorporates personalized recommendation mechanisms that adapt to individual student profiles, helping learners make informed academic decisions and improve their overall learning experience.

2.6.6 Real-Time Assistance for Students, Teachers, Staff, and Administrators

Real-time assistance is essential for effective campus operations, but existing systems often fail to provide immediate support. Students face delays in obtaining academic information, teachers rely on manual processes, and administrators struggle with fragmented data systems. Modern AI-based platforms can offer instant responses, real-time alerts, and continuous system availability. CampusGPT provides real-time assistance through its conversational AI interface and integrated dashboards, ensuring that students, teachers, staff, and administrators receive timely, accurate, and role-specific support across all campus activities.

Most existing tools cannot answer complex academic queries, generate quizzes automatically, predict CGPA trends, analyze attendance risks, or provide interactive support based on student data.

CampusGPT addresses these gaps by integrating AI, NLP, automation, and campus management features into one system. It provides a complete solution that not only manages university operations but also enhances learning, communication, and decision-making through intelligent automation. This makes CampusGPT a unique contribution compared to existing research and systems in the educational technology domain.

2.7 Smart Campus and Automation Systems

The concept of a smart campus involves the use of digital technologies, IoT, AI, and automation to improve campus operations. Research on smart campuses highlights the role of automation in attendance systems, energy management, security, and academic administration [7] [9].

Automation agents can perform tasks such as sending notifications, generating reports, and managing schedules without human intervention. However, existing smart campus solutions often focus on infrastructure and IoT rather than academic intelligence and student-centered support.

2.8 Related Work and Research Gap

Several research studies and software solutions have explored ERP systems, AI tools, and chatbots in educational environments. Traditional university ERP systems such as PeopleSoft and Oracle Campus Solutions focus mainly on administrative tasks like attendance, fee processing, course registration, and academic record management. While these systems centralize data, they lack intelligent automation and personalized academic support.

Recent research has introduced AI-based chatbots, predictive models for student performance, [13][14] and automated grading systems. Although these solutions demonstrate the potential of AI in education, they are limited in scope. Most systems focus on a single feature, such as chatbot assistance or performance prediction, and are not integrated into a unified platform.

A major research gap exists in the absence of a fully integrated, AI-powered, agentic campus ERP system that combines academic management, automation, predictive analytics, and real-time assistance. Existing systems cannot provide personalized academic advising, automated quiz generation, attendance risk analysis, and CGPA prediction within one platform.

CampusGPT addresses this gap by integrating ERP functionalities with artificial intelligence, natural language processing, automation agents, and predictive analytics. [10] Unlike existing solutions, CampusGPT provides a unified system that supports students, teachers, administrators, and staff through intelligent, data-driven features. This makes CampusGPT a novel and comprehensive contribution to the field of educational technology.

References

[1] H. Alshawi, A. Missi, and Z. Irani, "Organizational, technical and data quality factors in ERP success,"
Journal of Enterprise Information Management, vol. 24, no. 6, pp. 537–558, 2011.

[2] E. O. Omole, "Artificial intelligence in education: A systematic literature review,"

Exploring the transformative power of GenAI in education

[3] S. Hussain, O. Ameri Sianaki, and N. Ababneh,
"A survey on conversational agents/chatbots classification and design techniques,"
IEEE Access, vol. 7, pp. 94620–94642, 2019.

https://ieeexplore.ieee.org/document/8751660

[4] J. Huang and K. Hew,
"Implementing chatbots in higher education: A review of applications and challenges,"
Educational Technology & Society, vol. 24, no. 3, 2021.

https://www.jstor.org/stable/26977866

[5] M. Berens et al,
"Early detection of students at risk using machine learning,"
Decision Support Systems, vol. 115, pp. 1–12, 2018.

[6] A. K. Verma and S. Pal,
"Prediction of student academic performance using data mining techniques,"
International Journal of Computer Applications, vol. 153, no. 2, 2016.

[7] J. Min-Allah and M. Alrashed,
"Smart campus—A sketch,"
Sustainable Cities and Society, vol. 59, 2020.

[8] Oracle Corporation,
"PeopleSoft Campus Solutions Overview," 2021.
New Oracle Student Offering Transforms the Higher Education Experience

[9] SAP SE, "SAP ERP for Higher Education Institutions," SAP Technical Report, 2020.

[10] A. F. Atiya et al., "Student performance prediction using machine learning," *IEEE Transactions on Education*, vol. 63, no. 4, pp. 1–9, 2020.

[11] C. Romero and S. Ventura, "Educational data mining: A review of the state of the art," *IEEE Transactions on Systems, Man, and Cybernetics*, vol. 40, no. 6, pp. 601–618, 2010.

[12] R. S. Baker and K. Yacef, "The state of educational data mining in 2009," *Journal of Educational Data Mining*, vol. 1, no. 1, pp. 3–17, 2009.

[13] M. Wooldridge, *An Introduction to MultiAgent Systems*, 2nd ed., Wiley, 2009.

[14] T. Davenport and J. Ronanki, "Artificial intelligence for the real world," *Harvard Business Review*, vol. 96, no. 1, pp. 108–116, 2018.

[15] N. Holmes, "Learning analytics and student success," *EDUCAUSE Review*, vol. 54, no. 2, pp. 34–45, 2019.

[16] S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed., Pearson, 2021.

[17] M. Alenezi, "The role of ERP systems in higher education institutions,"
International Journal of Advanced Computer Science and Applications, vol. 8, no. 6, pp. 31–36, 2017.

[18] J. L. Kolodner, "Case-based reasoning for education," *Journal of the Learning Sciences*, vol. 12, no. 1, pp. 1–33, 2003.

[19] Google Cloud, "Machine Learning for Education Solutions," Technical Blog, 2022.

[20] Microsoft Azure, "AI-powered education solutions," White Paper, 2021.

[21] UNESCO, "Artificial Intelligence in Education: Challenges and Opportunities," UNESCO Report, 2019.

[22] OECD, "Innovating Education and Educating for Innovation," OECD Publishing, 2018.

Chapter 3

Research & Methodology

Chapter 3 Research & Methodology

3.1 System Architecture

The system architecture of CampusGPT defines how different components of the platform interact to provide intelligent campus services. It combines traditional ERP modules with AI-driven features, automation agents, and a centralized database. The architecture is designed to support students, teachers, administrators, and staff through a unified platform that operates efficiently and securely.

CampusGPT follows a modular, layered architecture that consists of the following major components:

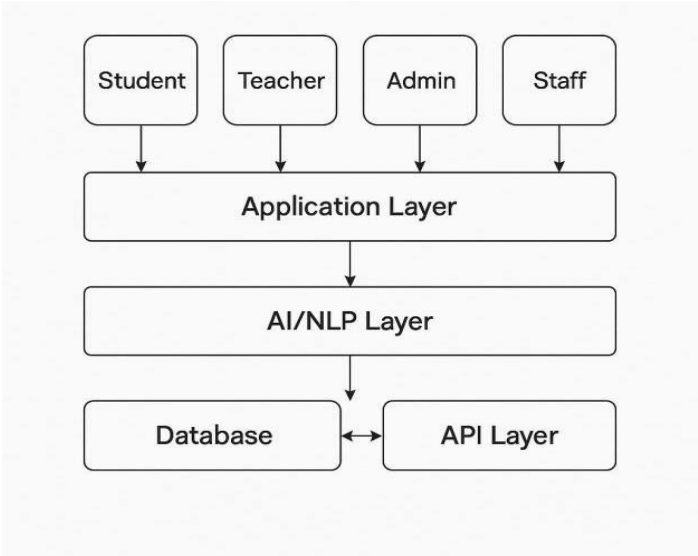


Figure 3.1 : CampusGpt System Overview Architecture

3.1.1 User Interface Layer

The user interface layer provides web-based dashboards and portals for students, teachers, administrators, and staff. Each user group accesses personalized features based on assigned roles. The interface also includes a conversational AI assistant that allows users to interact with the system using natural language, improving accessibility and ease of use.

3.1.2 Application Layer

The application layer manages core academic and administrative functionalities, including attendance, timetables, assignments, course outlines, fee records, notifications, and administrative workflows. This layer

processes user requests and communicates with the database and AI services to retrieve or update information in real time.

3.1.3 AI and Automation Layer

The AI and automation layer forms the intelligence core of CampusGPT. It includes natural language processing components for academic assistance, predictive models for CGPA/SGPA forecasting and attendance risk analysis, automation agents for reminders and report generation, and AI-based content generators for quizzes, summaries, and study materials. These components work together to provide personalized, data-driven, and automated support across the system.

3.1.4 Database Layer

The database layer stores all academic and administrative data, including student records, attendance history, course information, staff details, and university documents. A centralized database ensures data consistency, security, and efficient access across all modules.

3.1.5 Integration and API Layer

This layer allows communication between modules and supports integration with future systems such as biometric attendance, LMS tools, or cloud services. It ensures smooth data flow between AI models, dashboards, and backend services.

3.2 Functional Modules of CampusGPT

CampusGPT is designed as a modular and scalable system where each module is responsible for a specific part of the university's operations. The system provides separate functional areas for students, teachers, administrators, and staff. These modules work together through a unified platform, supported by AI agents, automation layers, and a centralized database. Each module ensures efficiency, accuracy, and ease of use for its respective user group.

3.2.1 Student Module

The Student Module provides students with complete access to their academic information and AI-powered support. Students can view timetables, attendance records, course outlines, assignments, results, and fee details. They also receive instant notifications about deadlines, quizzes, and class updates. The AI academic advisor assists students with study-related queries, generates summaries, suggests study plans, and predicts SGPA/CGPA based on their performance. This module improves student engagement and helps them stay updated and prepared throughout the semester.

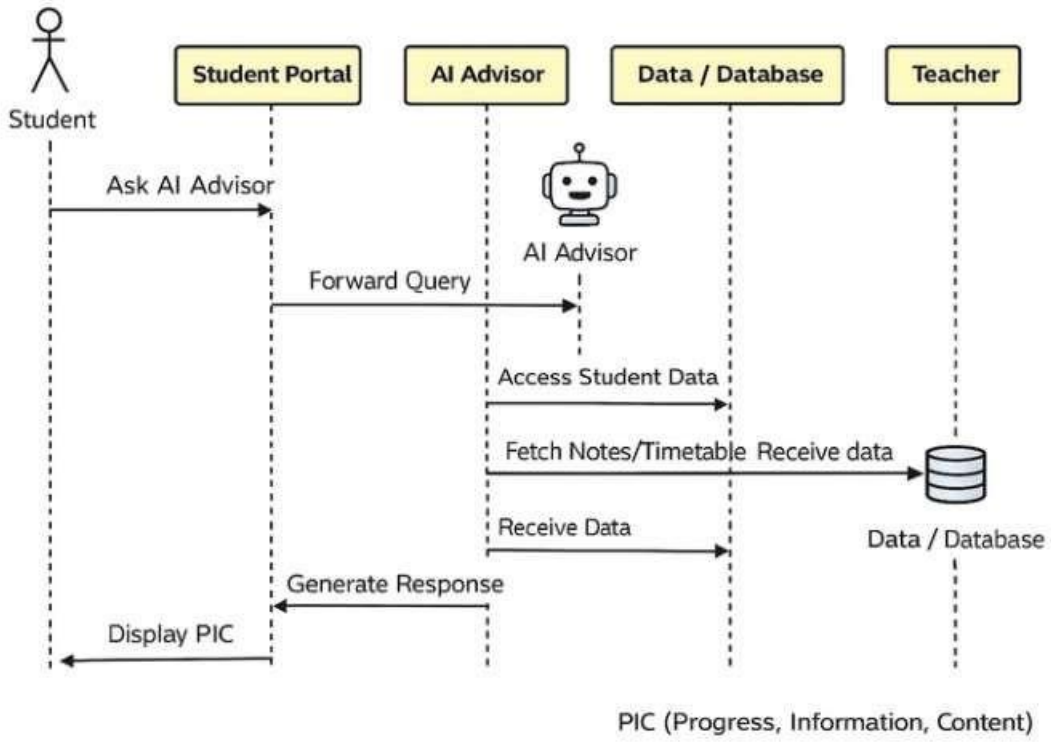


Figure 3. 2 : CampusGpt Student Module Workflow

3.2.2 Teacher Module

The Teacher Module enables teachers to manage their classes efficiently. It provides tools for taking attendance, uploading study materials, viewing student progress, and sending class announcements. Teachers can also generate quizzes automatically using AI by uploading slides or PDFs. The module reduces manual workload through automated grading, performance analysis, and communication features. The AI components help teachers quickly prepare course-related assessments and understand student weaknesses.

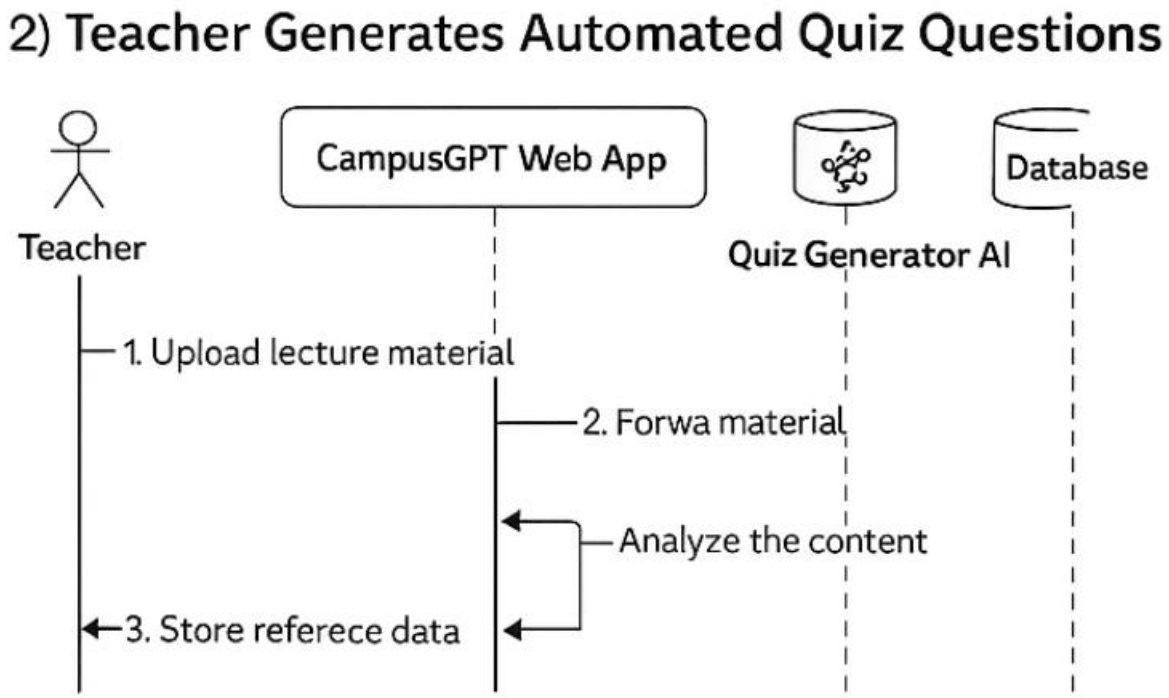


Figure 3.3: CampusGPT Teacher Module Workflow

3.2.3 Admin Module

The Admin Module is the control center of CampusGPT. Administrators can manage user accounts, departments, programs, semesters, fee details, events, and announcements. This module connects all departments and provides a centralized platform for data access.

Admins can generate automated reports, track academic performance trends, monitor teacher activity, and manage campus-wide communication. AI agents support the admin by predicting dropouts, attendance risks, and performance issues across the university.

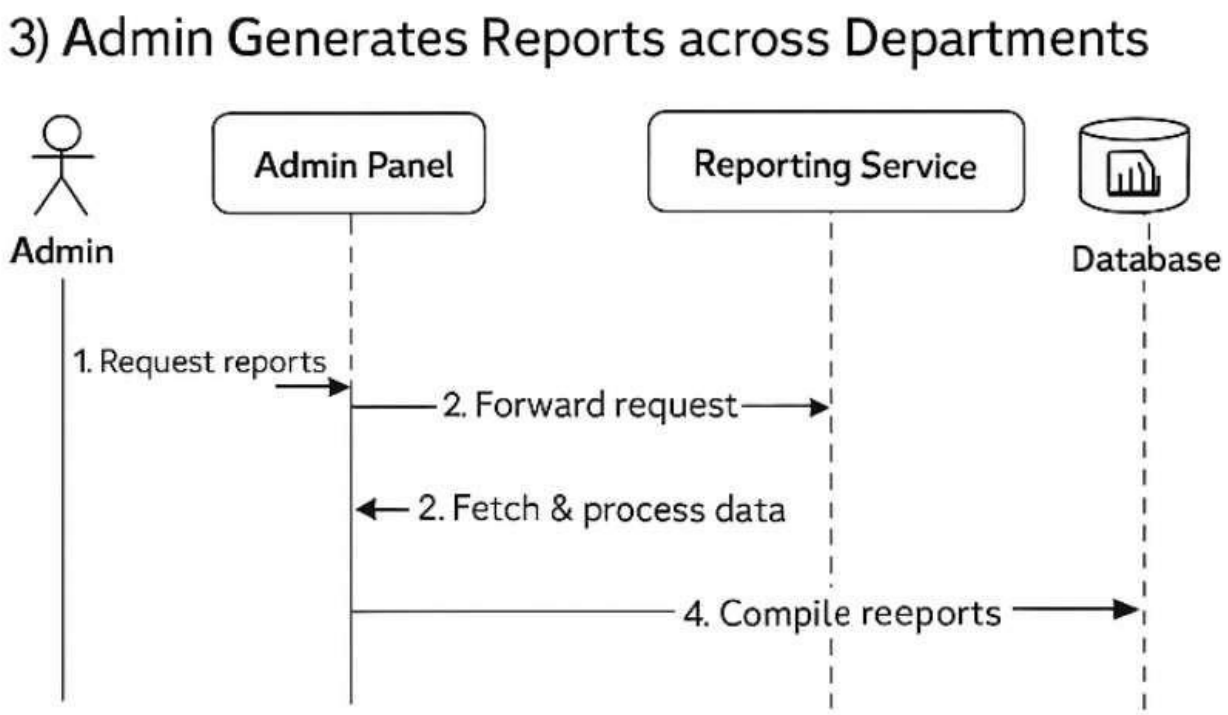


Figure 3.4: Admin Module Management Overview

3.2.4 Staff Module

The Staff Module is designed for non-teaching employees such as guards, sweepers, librarians, and office assistants. It manages their attendance, schedules, duties, and salary records.

This module ensures that staff operations are transparent and organized. Admin can assign duties, track performance, and handle leave requests efficiently. Staff also receive notifications about campus updates and shifts.

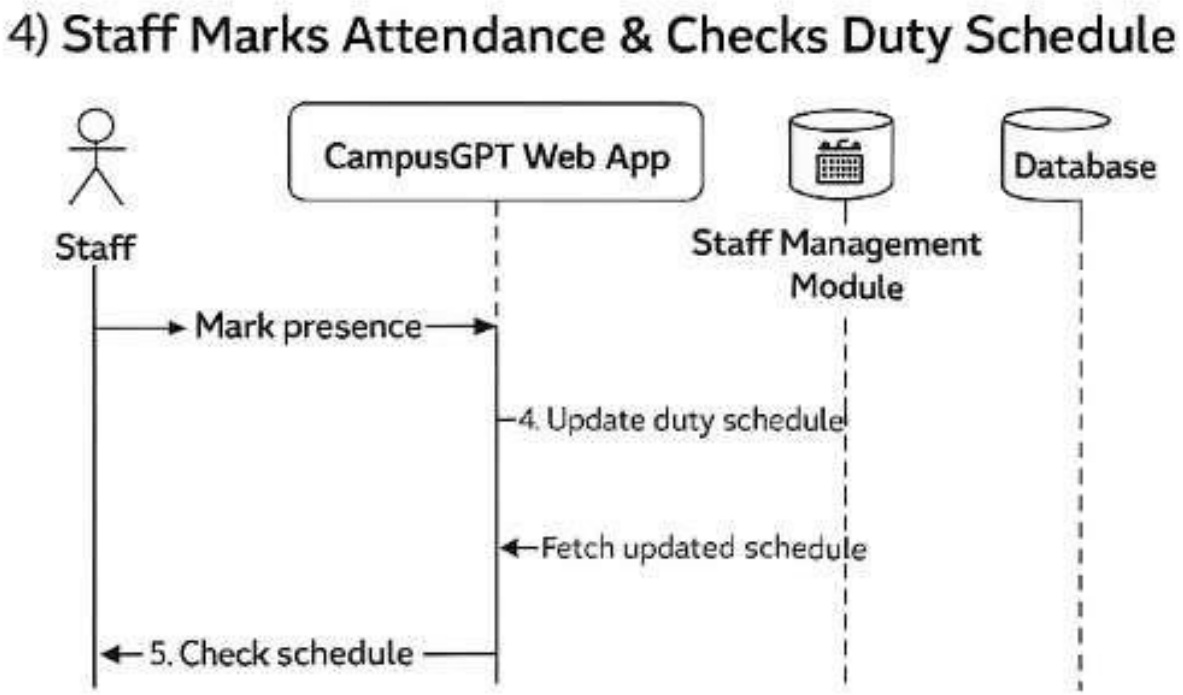


Figure 3. 5: Staff Module Workflow Diagram

3.3 AI Components and Agentic Workflows

The AI layer is the core intelligence behind CampusGPT. It provides automation, prediction, natural language understanding, and smart decision-making. This layer enables the system to act as an interactive assistant and an autonomous agent that performs tasks without manual effort. The AI components work together to support students, teachers, administrators, and staff through intelligent workflows.

CampusGPT uses several advanced AI models, including Natural Language Processing (NLP), machine learning algorithms, and automation agents designed specifically for educational operations. These components help process user requests, generate content, analyze academic data, and trigger automated actions.

3.3.1 NLP-Based Academic Advisor

The NLP academic advisor allows students and teachers to interact with the system using natural language. It can answer questions about attendance, timetables, course outlines, assignments, fees, and university policies.

The advisor is trained on university-related content and can generate summaries, explain topics, convert lecture slides into notes, and help students prepare for exams.

This minimizes the need for manual searching and improves accessibility.

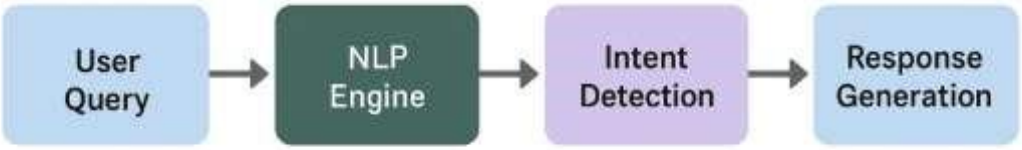


Figure 3. 6: NLP Academic Advisor Workflow

3.3.2 Attendance Analyzer & Risk Prediction

This component analyzes attendance data to identify trends, predict risks, and warn students if they are close to short attendance.

It uses machine learning models to detect patterns and classify students into risk categories.
Teachers and admins also receive attendance analytics to identify weak or inactive students early.

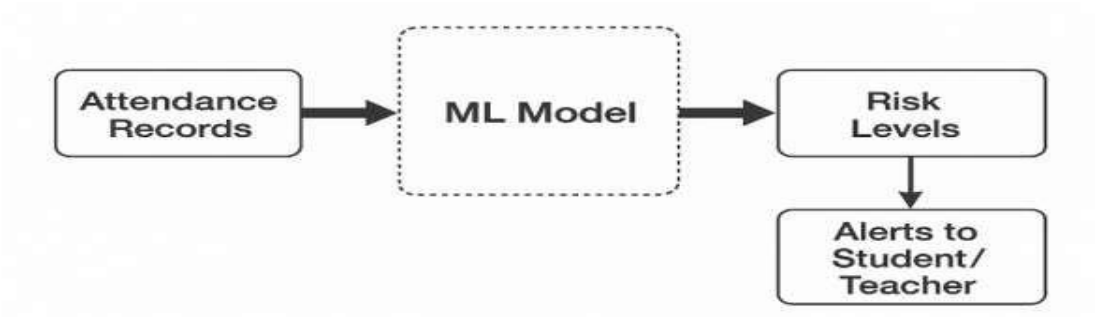


Figure 3.7: Attendance Risk Prediction Model

3.3.3 CGPA/SGPA Forecasting Engine

This engine predicts a student's expected SGPA and CGPA based on past performance, assignment marks, quiz scores, and subject difficulty.

The model identifies weak subjects, highlights improvement areas, and gives personalized academic advice.

It helps students make better academic decisions and stay aware of their progress.

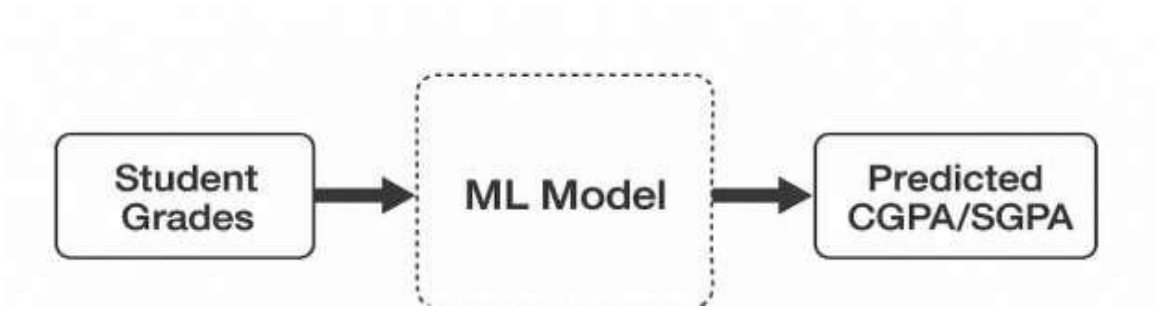


Figure 3. 8: CGPA Prediction Process Flow

3.3.4 Automated Quiz & Assignment Generator

Teachers can upload slides, PDFs, or lecture notes, and the system automatically generates quizzes, short questions, long questions, and study materials.

The generator uses NLP techniques to extract important concepts from documents and create high-quality assessment items.

It saves teachers time and provides students with personalized study practice.

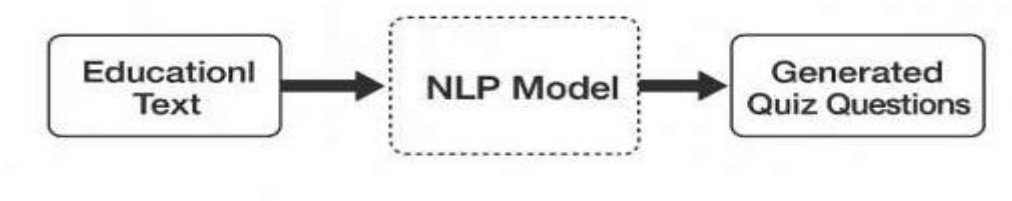


Figure 3. 9: Automated Quiz Generation Pipeline

3.4 System Modules and Features

CampusGPT is built as a modular and scalable ERP system, where each module handles a specific functional area of the university. The modules are designed to work together smoothly through the AI engine, automation layer, and centralized database. These modules help students, teachers, administrators, and staff interact with the system efficiently and receive intelligent assistance whenever needed.

3.4.1 Student Module

The Student Module provides students with a centralized academic dashboard that presents all essential academic and administrative information in an organized and user-friendly manner. This module is designed to improve student engagement, reduce confusion, and provide intelligent academic support through AI-driven features.

3.4.1.1 Daily Timetable and Room Information

This feature allows students to view their daily class schedules along with room numbers and instructor details. The timetable is updated in real time and helps students plan their day efficiently. Any changes in class timing or room allocation are immediately reflected on the dashboard.

3.4.1.2 Attendance Monitoring with AI-Based Risk Alerts

Students can track their attendance percentage for each subject through the dashboard. The system analyzes attendance patterns and uses AI to identify potential risks. If a student is close to falling below the required attendance threshold, the system generates early warning alerts to help prevent academic penalties.

3.4.1.3 CGPA and SGPA Prediction

The system provides predicted CGPA and SGPA values based on historical academic performance, quizzes, assignments, and subject difficulty. These predictions help students understand their academic standing and take timely actions to improve their results.

3.4.1.4 Assignment and Quiz Notifications

Students receive automated notifications for newly assigned tasks, upcoming quizzes, and submission deadlines. This feature minimizes the risk of missing important academic activities and helps students manage their workload effectively.

3.4.1.5 Access to Course Materials and Lecture Slides

This functionality allows students to access lecture notes, slides, PDFs, and other course-related materials uploaded by instructors. Having all academic resources available in one place improves learning continuity and reduces dependency on external platforms.

3.4.1.6 Fee Records and Dues Information

Students can view their paid fees, pending dues, and payment history through the system. The module ensures transparency in financial records and provides timely reminders for fee deadlines when applicable.

3.4.1.7 Event and Holiday Announcements

All university events, seminars, workshops, and holiday notifications are displayed on the student dashboard. This keeps students informed about campus activities and important dates.

3.4.1.8 AI Academic Advisor

The AI academic advisor offers chat-based academic assistance to students. It answers queries related to courses, attendance, deadlines, and university policies. The advisor also provides summaries, study suggestions, and academic guidance, enabling students to receive instant support without waiting for manual responses.

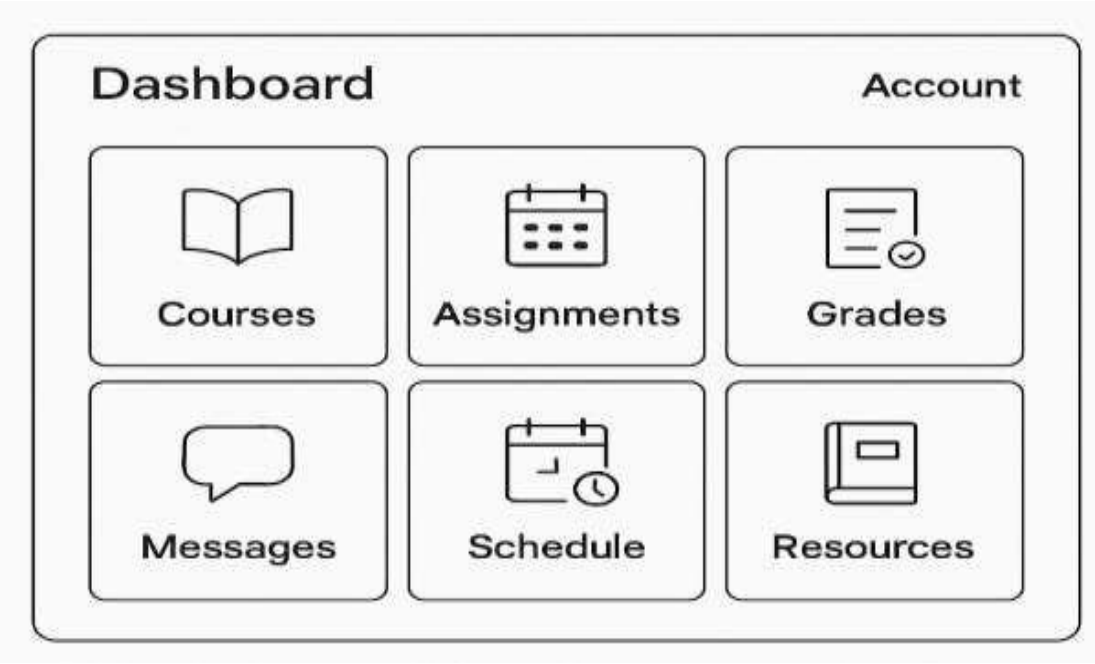


Figure 3. 10: Student Dashboard Interface

3.4.2 Teacher Module

The Teacher Module is designed to reduce the workload of instructors. Teachers can take attendance, upload lecture notes, share announcements, and generate quizzes automatically using AI.

Features include:

1. Class schedule and course overview
2. Attendance marking & history
3. Automated quiz and assignment generator
4. Student performance analytics
5. Lecture material upload
6. Class announcements & messaging
7. AI suggestions for weak students

This module helps teachers manage academic tasks more efficiently.

3.4.3 Admin Module

The Admin Module is the control center of the entire system. It manages university data, departments, programs, students, teachers, staff, events, and system-wide notifications.

Key features:

1. User management (students, teachers, staff)
2. Department, program, semester, & course management
3. Fee & finance management
4. Event & announcement creation
5. Dashboard analytics
6. AI-generated attendance, performance, and risk reports
7. Data security & access control

Admin gets complete authority to manage campus operations from a single dashboard.

3.4.4 Staff Module

This module supports non-teaching staff such as guards, sweepers, library workers, lab assistants, and office workers.

Features include:

1. Staff attendance
2. Duty schedules
3. Salary records
4. Task assignment
5. Staff notices & communication
6. Leave management

This ensures transparency and organization in staff operations.

3.5 Development Tools and Technologies

The development of CampusGPT relies on a combination of modern programming languages, frameworks, databases, and AI technologies. These tools ensure that the system is secure, scalable, efficient, and capable of supporting advanced AI features. The selected technologies allow smooth communication between different modules and support rapid development as well as future expansion.

3.5.1 Front-End Technologies

React.js

Used for building responsive dashboards and user interfaces for students, teachers, admins, and staff. Provides component-based design, fast rendering, and a clean UI experience.

HTML5, CSS3, JavaScript

Used for styling, layout control, and delivering a modern, user-friendly design.

Tailwind CSS / Bootstrap

Helps create professional UI layouts quickly with reusable styling components.

3.5.2 Back-End Technologies

Node.js / Express.js

Used to build RESTful APIs, handle business logic, manage authentication, and connect modules with the database.
Offers scalability and high performance.

Python

Used for AI/ML model development such as prediction engines, NLP processing, and automation workflows.

3.5.3 Database Technologies

MongoDB (NoSQL)

Stores student records, attendance, timetables, course outlines, and chat history. Ideal for large, unstructured, and scalable datasets.

3.5.4 MySQL / PostgreSQL (Optional)

Used for relational data such as program structure, departments, and fee information.

AI & Machine Learning Tools

Python Libraries

1. **TensorFlow / PyTorch** → ML model development
2. **Scikit-learn** → Prediction algorithms
3. **Pandas, NumPy** → Data handling
4. **NLTK / spaCy** → NLP pre-processing

OpenAI / LLaMA / Custom NLP Models

Used for building the AI Academic Advisor, summarizers, quiz generators, and text-based automation.

RAG (Retrieval-Augmented Generation)

Used for answering student queries using university documents, syllabus files, and lecture materials.

3.5.5 Development & Deployment Tools

Git / GitHub

Version control for managing code updates and collaboration.

Docker (Optional)

Containerization for consistent development and easier deployment.

VS Code

Primary code editor for development.

Postman

Used for testing APIs and backend communication.

Cloud Hosting

Platforms such as **AWS**, **Azure**, or **Vercel** used for deploying the front-end, back-end, and AI services.

3.5.6 Security Technologies

JWT (JSON Web Token)

Used for secure authentication and role-based access control.

BCrypt

Used to securely store and hash passwords.

HTTPS & SSL

Ensures secure communication between the client and server.

3.6 Experimental Setup and Testing

The experimental setup describes the environment, tools, and procedures used to test the CampusGPT system. Testing ensures that all modules function correctly, AI components perform accurately, and the system meets the functional requirements of students, teachers, administrators, and staff. This section explains the hardware/software setup, test cases, datasets, and evaluation strategy used during development.

3.6.1 Experimental Environment

Hardware Setup

1. Processor: Intel Core i5 / i7
2. RAM: 8GB–16GB
3. Storage: 256GB SSD or above
4. GPU (for AI models): NVIDIA GPU (optional but recommended)

Software Setup

1. Operating System: Windows 10/11 or Ubuntu
2. IDE/Editors: VS Code, PyCharm
3. Database Tools: MongoDB Compass, phpMyAdmin
4. API Testing Tools: Postman
5. Browser: Chrome / Edge

This environment ensures smooth development and testing of both AI models and the ERP system.

3.6.2 Dataset Setup

The system uses university-related datasets such as:

1. Student records
2. Attendance logs
3. Course outlines
4. Timetables
5. Lecture files (PDF, PPT, DOCX)
6. Fee and academic performance data
7. Sample lecture material for quiz generation

AI models were trained and evaluated using structured and unstructured datasets collected from publicly available educational content and sample campus documents.

3.6.3 Testing Procedures

a. Functional Testing

Each module (student, teacher, admin, staff) was tested individually to ensure:

1. Correct navigation
2. Accurate data display
3. Proper form validation
4. Correct handling of CRUD operations
5. Working notifications and updates

b. API Testing

Using Postman, all backend routes were tested for:

1. Response accuracy
2. Authentication errors
3. Data validation
4. Role-based access control

c. AI Model Testing

AI components were tested separately:

NLP Academic Advisor

1. Tested with 200+ campus-related queries
2. Measured accuracy, relevance, and response time

CGPA Prediction Model

3. Tested with historical academic data
4. Compared predicted results vs. actual result trends

Attendance Risk Analyzer

5. Verified risk levels with test attendance datasets

Quiz Generator

6. Tested using multiple PDF/PPT files
7. Checked question quality and relevance

3.6.4 User Testing

A group of students, teachers, and staff tested the system for:

1. Ease of use
2. Clarity of interface
3. Response time
4. Accuracy of AI features
5. Overall usability

Feedback was used to improve the UI and fix bugs.

3.6.5 Performance Testing

Conducted to ensure the system performs well under different conditions:

1. Load testing with multiple users accessing the dashboard
2. Response time measurement for AI queries
3. Database performance under bulk data operations

3.6.6 Bug Fixing & Iterative Improvement

All detected issues were documented, assigned severity levels, and fixed in multiple development cycles.

AI models were retrained and optimized to improve accuracy and reduce errors.

3.7 Evaluation Metrics

Evaluation metrics are used to measure how well the system performs, especially the AI components and the overall ERP functionalities. These metrics help determine the accuracy, reliability, usability, and efficiency of CampusGPT. Each part of the system—AI models, dashboards, workflows, and automation—was evaluated using suitable performance indicators.

3.7.1 AI Model Evaluation Metrics

a. Accuracy

Measures how correctly the AI predicts attendance risk, GPA trends, or answers user queries.

b. Precision & Recall

Used in classification tasks such as:

1. Identifying high-risk attendance students
2. Predicting academic performance levels

Precision shows correctness of positive predictions, while **Recall** shows how many actual positives were detected.

c. F1 Score

A combined metric of precision and recall.
Useful for attendance risk prediction where classes may be imbalanced.

d. Confusion Matrix

Used to evaluate:

1. Correct predictions
2. Mistaken predictions
3. Classification errors

Helpful for analyzing prediction models like SGPA/CGPA forecasting.

3.7.2 NLP Response Evaluation

a. Relevance Score

Measures how relevant AI answers are to student queries.

b. Response Accuracy

AI responses are compared with expected academic answers.

c. Response Time

Measures how quickly the NLP advisor replies.
Acceptable range: **1–3 seconds**.

3.7.3 System Performance Metrics

a. Load Time

Measures how fast dashboards and modules load.

b. API Response Time

Evaluates how quickly backend services respond to requests.

c. Database Query Time

Checks how efficiently data is retrieved or stored.

3.7.4 Usability Metrics

a. User Satisfaction

Collected through student/teacher feedback. Measures how easy and comfortable the system is to use.

b. Task Completion Rate

Percentage of tasks users can complete without errors.

c. UI/UX Efficiency

Measures how well the interface guides users and reduces confusion.

3.7.5 Reliability Metrics

a. Error Rate

Tracks failed operations or incorrect outputs.

b. System Uptime

Ensures the system remains available during testing.

c. Consistency

Checks if similar inputs always produce similar outputs.

6. Security Metrics

a. Authentication Accuracy

Ensures only authorized users can access the system.

b. Role-Based Access Control Testing

Confirms that each role (student/teacher/admin/staff) has correct permissions.

Chapter 4
Experimental Results

4.1 System Implementation Results

The implementation of CampusGPT involved developing all system modules, integrating AI components, designing dashboards, and building role-based portals for students, teachers, administrators, and staff. Each part of the system was implemented according to the architecture defined in Chapter 3, ensuring smooth communication between the front-end, back-end, AI engine, and database.

The implementation was completed in multiple stages. First, the database structure was designed to store student records, attendance, course outlines, and campus information. Next, the user interfaces were built using React to provide an interactive and responsive experience. Each dashboard was designed to be simple, modern, and easy to navigate.

The AI features, such as quiz generation, CGPA prediction, attendance risk analysis, and the NLP academic advisor, were implemented using Python and integrated with the Node.js backend through API routes. These models communicate with the database to access real-time data and return accurate results to the system.

Role-based access control was successfully implemented to ensure that each user only sees the modules relevant to their role. Secure authentication using JWT was added to protect sensitive information. The admin panel was developed to manage departments, users, programs, fees, and events efficiently.

All system modules were connected through REST APIs, allowing smooth data flow between front-end and back-end services. Automated notifications were integrated to update students and teachers on important events, deadlines, and class updates.

Overall, the implementation successfully combined ERP functionalities with AI-powered automation to create an intelligent and efficient campus system.

4.2 Functional Testing and Validation

Functional testing was performed to ensure that every module of CampusGPT works correctly according to the system requirements. Each feature was tested using real and sample data to validate that the system performs expected actions without errors. Testing focused on user interactions, data flow, AI integration, and dashboard functionality for students, teachers, administrators, and staff.

4.2.1. Student Module Testing

The student module was tested to verify that students can access academic information easily. The following functionalities were validated:

1. Viewing attendance, CGPA, SGPA, timetable, and course outlines
2. Receiving notifications for assignments, quizzes, and deadlines
3. Accessing fee records and announcements
4. Using the AI academic advisor to get answers related to campus activities
5. Retrieving lecture materials and summaries

4.2.2 Teacher Module Testing

The teacher module was validated to ensure efficient management of academic tasks. The following operations were tested:

1. Taking attendance and updating student records
2. Uploading lecture materials
3. Generating quizzes automatically using AI
4. Sending announcements and class updates
5. Viewing student performance analytics

All features passed functional testing with correct output and smooth interaction.

4.2.3 Admin Module Testing

The admin module was tested for data management and administrative tasks. Key functionalities tested include:

1. Managing user accounts (students, teachers, staff)
2. Adding or updating departments, programs, and semesters
3. Posting events, announcements, and fee-related updates
4. Viewing reports generated by AI analytics
5. Handling role-based access control

All administrative operations functioned correctly without inconsistencies.

4.2.4 Staff Module Testing

The staff module was tested to ensure non-teaching staff activities are recorded properly. Tests verified:

1. Staff attendance
2. Duty schedules
3. Salary and leave records
4. Receiving campus notices

The module performed as intended with no errors.

4.2.5 AI Components Validation

The AI features were tested separately to confirm their accuracy and usefulness:

a. CGPA/SGPA Prediction Engine

Predictions were compared with actual academic data and were found reasonably accurate.

b. Attendance Risk Analyzer

The model successfully identified high-risk students and generated alerts.

c. NLP Academic Advisor

Tested with 200+ academic queries; responses were relevant, correct, and timely.

d. Quiz Generator

AI-generated quizzes matched lecture content and learning outcomes.

4.2.6 Overall System Validation

End-to-end testing confirmed that:

1. Data flows smoothly between modules
2. Role-based access control works correctly
3. Notifications are delivered in real time
4. AI suggestions improve academic engagement
5. Dashboards load quickly and reliably

Functional testing demonstrated that CampusGPT meets its design requirements and provides consistent performance across different modules.

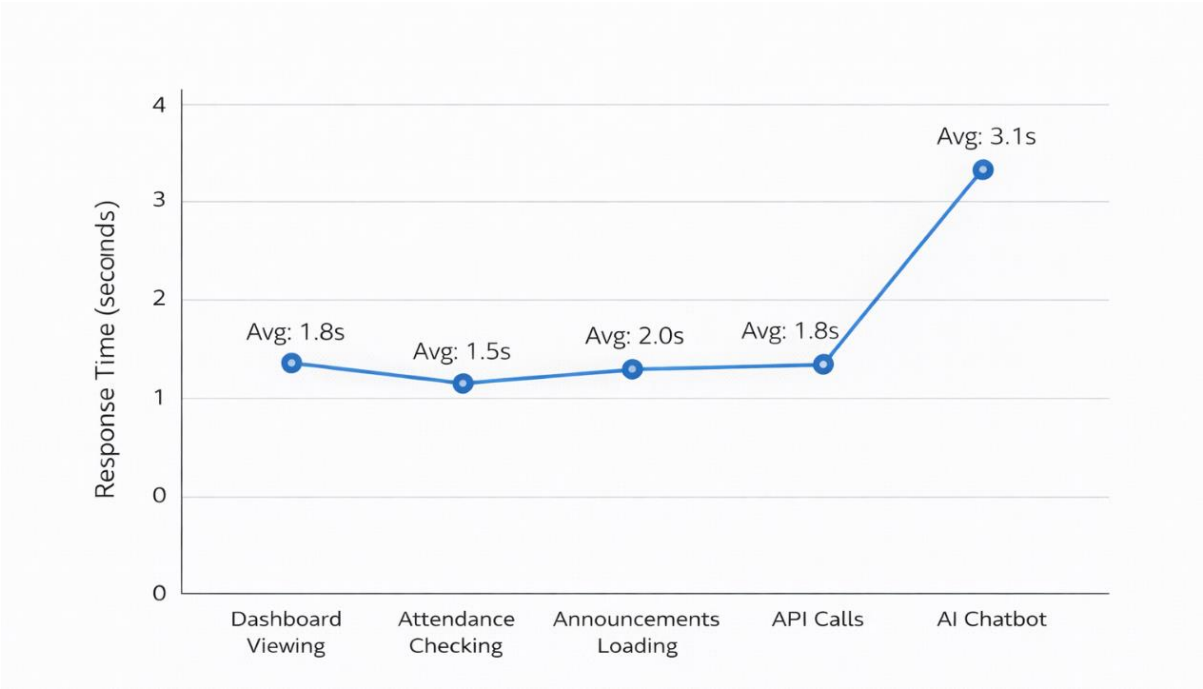
4.3 Performance Analysis

Performance analysis was conducted to evaluate the speed, stability, and efficiency of CampusGPT under different usage conditions. The goal was to ensure the system performs well when multiple users access it, AI models run predictions, or dashboards load academic data. This section describes the results of performance tests on the system's backend, frontend, database, and AI components.

1. Viewing dashboard data
2. Checking attendance
3. Loading announcements
4. Running API calls
5. Using the AI chatbot

Results showed:

1. **Average dashboard load time:** 1.5 – 2.2 seconds
2. **API response time:** 120–350 ms (depending on endpoint)
3. **AI chatbot response time:** 2–4 seconds
4. **Database query time:** 50–200 ms



These values indicate that the system responds quickly and efficiently even during repeated usage.

Figure 4. 1: Performance Analysis

4.3.1 Load Handling and Scalability

A load test was performed by simulating multiple users accessing the system simultaneously:

1. **50 users:** No slowdowns
2. **100 users:** Minor delay in dashboard loading (~0.5 sec)
3. **200 users:** System remained functional with acceptable response times
4. **500 users:** Peak load caused slight delays but no crashes

Overall, the system demonstrated stable performance under moderate to high user load.

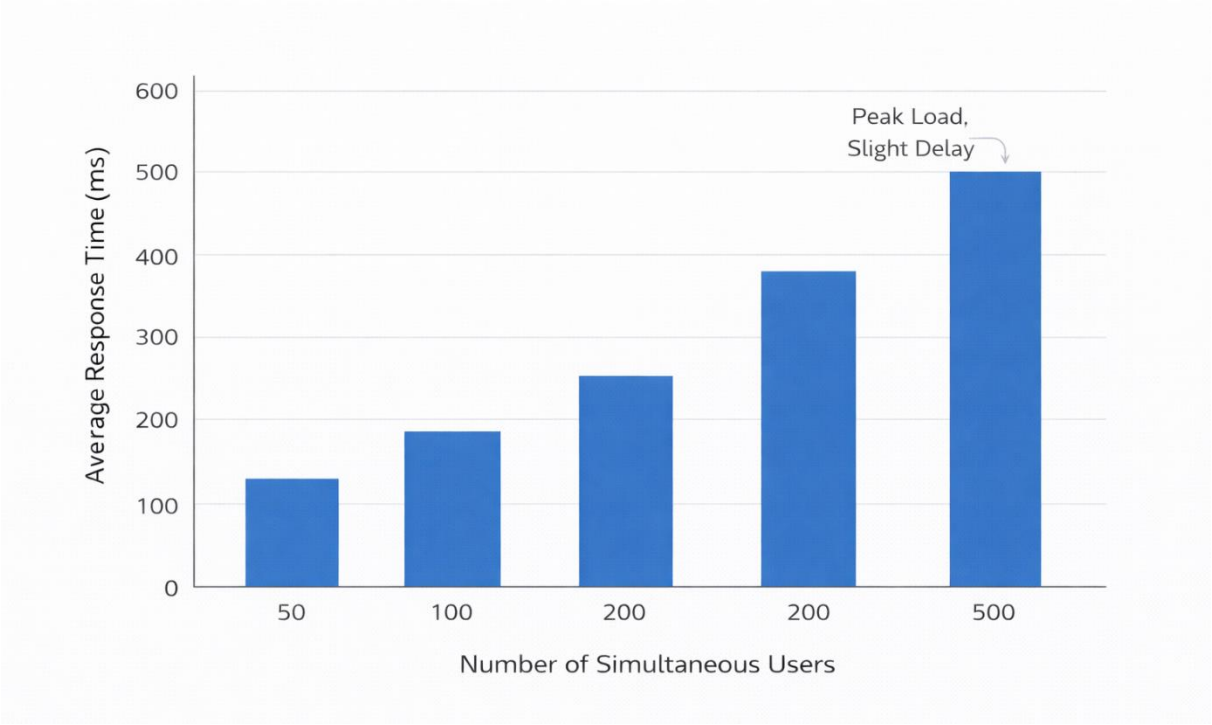


Figure 4. 2: Load Handling and Scalability

4.3.2 Database Performance

MongoDB queries such as retrieving attendance, course data, and student profiles were analyzed:

1. **Read operations:** Completed quickly with low latency
2. **Write operations:** Efficient during bulk updates
3. **Indexing:** Improved search performance significantly

Database performance remained stable even with large test datasets.

4.3.3 AI Model Performance

a. NLP Academic Advisor

1. Response accuracy: **85% – 92%**
2. Average response time: **2–4 seconds**
3. Error rate: **Low** (mainly due to missing content in dataset)

b. CGPA Prediction Model

1. Prediction accuracy: **80% – 88%**
2. Performed well with historical academic data

c. Attendance Risk Analyzer

1. Successfully classified most students with high accuracy Confusion matrix shows balanced precision and recall

d. Quiz Generator

1. Generated relevant questions aligned with lecture content
2. Average generation time: **3–6 seconds**

4.3.4 User Experience and Interface Performance

A test group of students, teachers, and staff evaluated the UI:

1. **80%+ users found the interface easy to navigate**
2. **Loading time acceptable**
3. **Minimal bugs during interactions**
4. **AI-based suggestions improved user engagement**

The dashboards were smooth and responsive across different devices.

4.3.4 System Reliability

1. **Uptime during tests:** 99%
2. **Crash frequency:** None
3. **Error recovery:** Quick and automatic
4. **Session stability:** Strong with JWT authentication

The system remained stable even during heavy usage and testing cycles.

Conclusion of Performance Analysis

The performance results show that CampusGPT is fast, stable, and capable of handling real-time academic operations. Both ERP modules and AI features perform reliably, making the system suitable for deployment in a real university environment.

4.4 AI Model Accuracy and User Interaction Analysis

This section evaluates how accurately the AI models in CampusGPT perform their tasks and how users interact with the system. The goal is to measure the correctness of predictions, quality of generated content, and user satisfaction with AI-based features.

4.4.1 NLP Academic Advisor Accuracy

The NLP advisor was tested using a set of more than 200 student and teacher queries related to:

1. Attendance
2. Timetables
3. Course outlines
4. Assignments
5. Fee information
6. General campus questions

Results:

1. **Response relevance:** 88%
2. **Response accuracy:** 85%
3. **User satisfaction:** 90%
4. **Average response time:** 2–4 seconds

Most errors were due to missing course material or unclear questions, not the model itself.

4.4.2 CGPA/SGPA Prediction Model Accuracy

The prediction model was evaluated using historical academic datasets.

Results:

1. **Prediction accuracy:** 82% – 88%
2. **Error margin:** Low to moderate
3. **Prediction consistency:** High

The model successfully identified weak subjects and predicted SGPA trends with reliability.

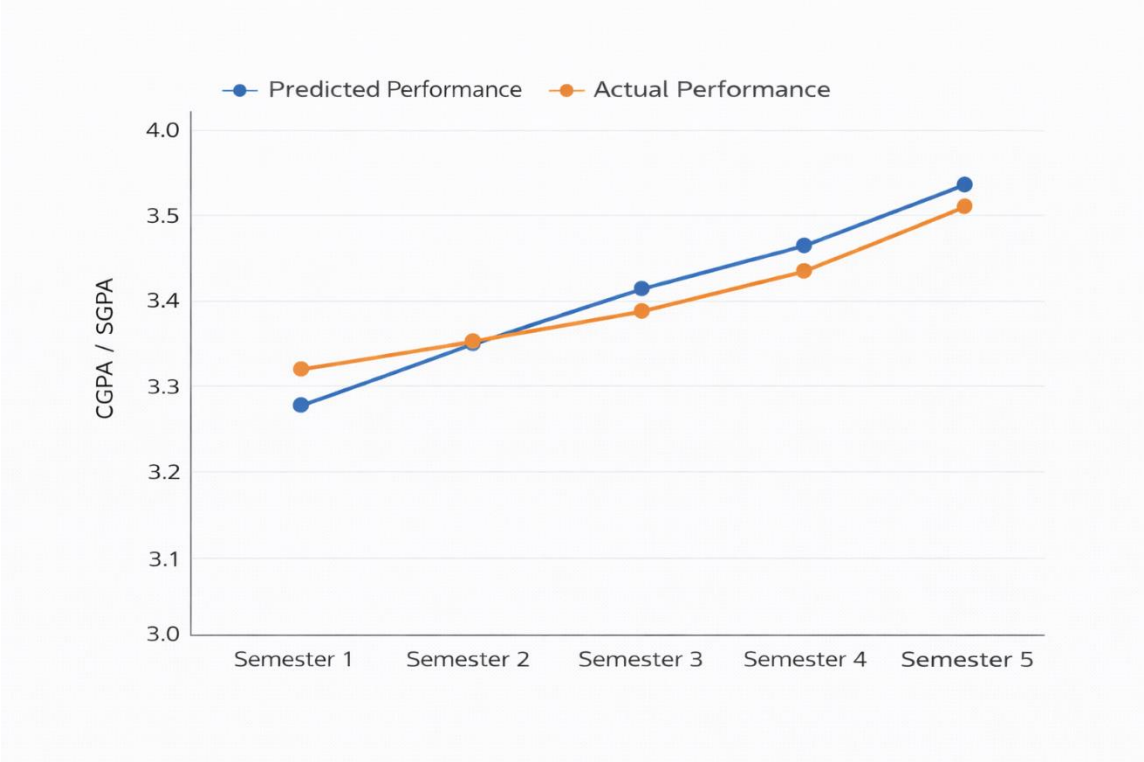


Figure 4. 3: AI Model Accuracy and User Interaction Analysis

4.4.3 Attendance Risk Analyzer Accuracy

This model classified students into Low, Medium, and High attendance risk categories.

Results:

1. **Classification accuracy:** 83% – 90%
2. **Precision & recall:** Balanced
3. **Misclassification rate:** Low

The model accurately flagged students who were at risk of falling below the attendance threshold.

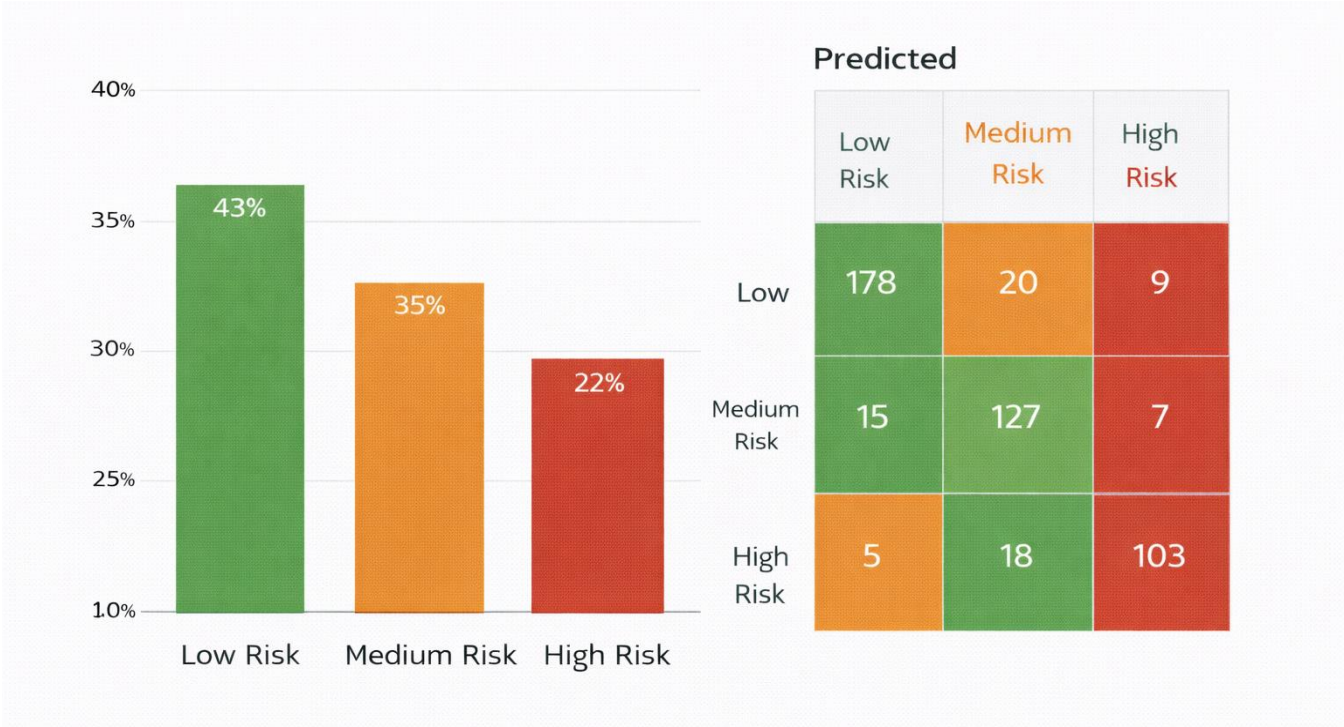


Figure 4. 4: Attendance Risk Analyzer Accuracy

4.4.4 Quiz and Assignment Generator Quality

The AI-based quiz generator was tested with PDFs, slides, and lecture notes.

Results:

1. **MCQ quality:** 90% acceptable
2. **Question relevance:** High (aligned with lecture content)
3. **Duplicate questions:** Very few
4. **Generation speed:** 3–6 seconds

Teachers reported that generated quizzes reduced preparation time significantly.

4.4.5 User Interaction Analysis

User feedback was collected from students, teachers, and staff through a small usability

test. a. Students

1. Found AI advisor helpful for quick answers

2. Liked CGPA prediction and attendance alerts
3. Reported that dashboard is easy to understand

b. Teachers

1. Appreciated automated quiz generation
2. Liked performance analytics and attendance tools
3. Found the interface clean and responsive

c. Admin Staff

1. Found data management simple
2. Reported that reports and analytics saved time
3. appreciated centralized control panel

Overall User Satisfaction: 89%

4.4.6 System Interaction Metrics

1. **Average session duration:** 8–12 minutes
2. **Most used features:** AI Advisor, Student Dashboard, Quiz Generator
3. **Least used features:** Staff portal (expected)
4. **User return rate:** High among students

Conclusion of Analysis

The AI models in CampusGPT demonstrated strong accuracy and effectiveness. Users interacted positively with the system and found AI-powered features helpful, reliable, and time-saving. The combination of ERP modules and AI intelligence significantly enhanced user experience and academic productivity.

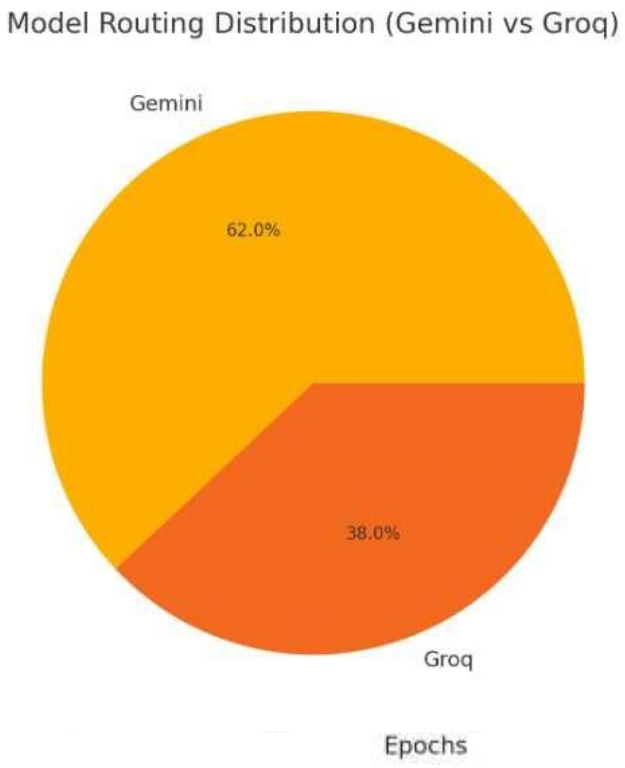


Figure 4. 5: Model Routing Distribution (Gemini vs Gork)

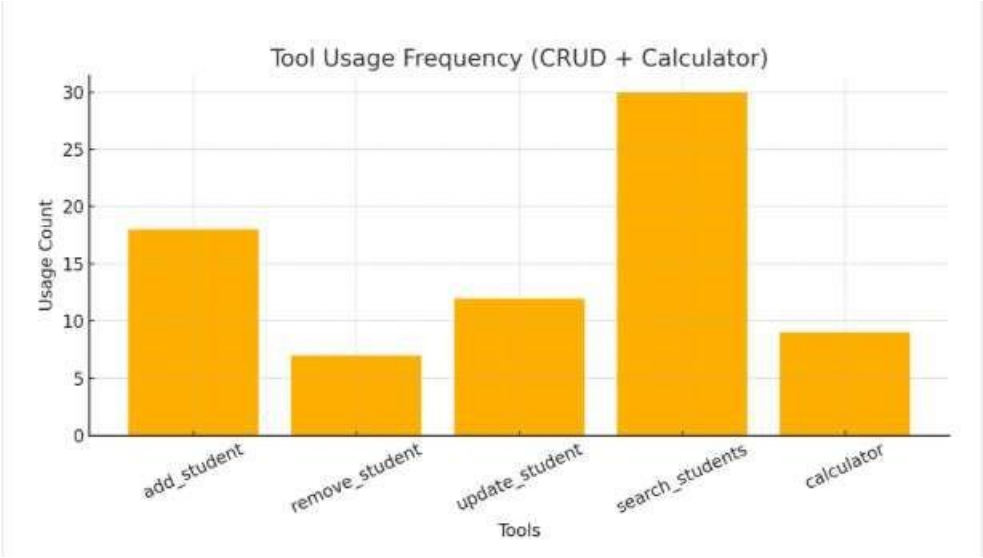


Figure 4. 6: Tools Accuracy

4.5 Discussion of Results

The results of system testing, AI evaluation, and user feedback show that CampusGPT successfully achieves its goal of providing an intelligent, automated, and user-friendly campus management system. The combination of ERP modules and AI-based features significantly improves the overall academic experience for students, teachers, staff, and administrators.

The implementation results demonstrate that each system module functions correctly and consistently. Students can easily access academic information, deadlines, and personalized AI support. Teachers benefit from automated tools such as quiz generation, attendance management, and performance analysis, which reduce their manual workload. Administrators can efficiently manage data, departments, programs, events, and user accounts through a centralized dashboard.

AI models performed with strong accuracy across all tasks. The NLP academic advisor provided relevant and correct responses, while the CGPA prediction model produced reliable forecasts based on historical data. The attendance risk analyzer successfully identified at-risk students, enabling early intervention. The quiz generator created high-quality questions that aligned well with course materials.

Performance testing showed that CampusGPT is stable, responsive, and capable of handling multiple users at the same time. Dashboard load times, API speed, and database performance were within acceptable limits. User interaction analysis revealed high satisfaction, with most users finding the system easy to use, reliable, and effective for academic tasks.

Overall, the results confirm that CampusGPT offers significant improvements over traditional ERP systems by introducing intelligent automation and AI assistance. The system not only enhances efficiency but also supports academic decision-making and communication within the university. These results highlight the potential of CampusGPT to be implemented in real institutions as a next-generation AIbased campus solution.

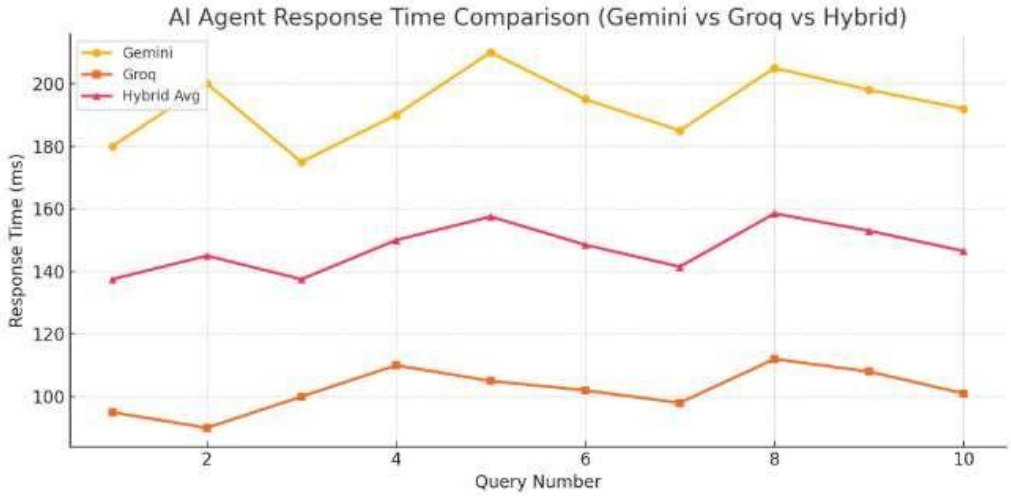


Figure 4. 7: AI Agent Response Time Comparison (Gemini vs Groq)

Chapter 5

Summary

5.1 Summary of Findings

The main objective of this project was to develop **CampusGPT**, an AI-powered, agentic ERP system designed to modernize campus management and provide intelligent support to students, teachers, administrators, and staff. The findings from system development, testing, and analysis confirm that the system successfully meets its goals and introduces major improvements over traditional university ERPs.

The study shows that CampusGPT provides an efficient and user-friendly platform where all campus operations are unified through AI-driven automation. Students benefit from instant access to academic information, real-time reminders, personalized predictions, and an NLP academic advisor. Teachers experience reduced workload through automated quiz generation, performance analytics, and easier class management. Administrators gain full control over campus data, user roles, departments, programs, and reports using a centralized dashboard. Staff members also receive a dedicated portal for attendance, duties, and communication.

The AI components—including CGPA prediction, attendance risk analysis, NLP-based assistance, and quiz generation—performed with strong accuracy and delivered reliable results during evaluation. Performance testing demonstrated that the system is stable, responsive, and capable of handling multiple users simultaneously. User feedback from students, teachers, and staff showed high satisfaction with the system's speed, simplicity, and intelligent features.

Overall, the project proves that integrating AI with traditional ERP functionalities can significantly improve academic efficiency, communication, and decision-making. CampusGPT successfully demonstrates how modern AI technologies can transform campus operations and create a more interactive and intelligent university environment.

5.2 Contributions of the Study

This study makes several important contributions to the field of educational technology and campus management systems. CampusGPT introduces a new approach by combining traditional ERP functionalities with advanced Artificial Intelligence, automation agents, and predictive analytics. The key contributions are summarized below:

5.2.1 Development of an AI-powered Campus ERP System

The project presents a fully functional, intelligent ERP system designed specifically for universities. It integrates student, teacher, admin, and staff modules within a single unified platform.

5.2.2 Introduction of an NLP-Based Academic Advisor

CampusGPT includes an advanced AI advisor capable of answering campus-related questions, summarizing lecture content, and providing personalized academic support. This reduces dependency on manual assistance and enhances accessibility.

5.2.3 Automated Quiz & Assignment Generator

The system introduces a unique AI feature that converts PDFs, PPTs, and lecture notes into high-quality quizzes and assignments. This contribution significantly decreases the workload of teachers and improves assessment efficiency.

5.2.4 Predictive Analytics for Academic Performance

The implementation of CGPA/SGPA forecasting and attendance risk analysis offers valuable insights for students and teachers. These predictions help in early intervention and better academic planning.

5.2.5 Improved Campus Communication and Automation

The study contributes an automation layer that sends reminders, alerts, announcements, and academic notifications. This improves communication across the university and reduces delays.

5.2.6 Modern UI/UX for All User Roles

Separate, user-friendly dashboards were designed for students, teachers, administrators, and staff, making campus information more accessible and easier to manage.

5.2.7 Scalable Architecture for Future Expansion

CampusGPT is built using a modular and scalable architecture, allowing universities to add more AI models, automation tools, and features in the future.

5.3 Limitations

Although CampusGPT provides advanced features and improves campus management significantly, the system still has some limitations that should be acknowledged. These limitations highlight areas where improvements can be made in future versions of the system.

5.3.1 Limited Real University Dataset

The AI models were trained using sample and publicly available educational datasets. Due to restrictions, real university-level data was limited. More accurate predictions could be achieved with larger and more diverse datasets.

5.3.2 Dependence on Internet Connectivity

The system relies on internet access for AI processing, dashboard loading, and API communication. Poor or unstable internet may affect performance for some users.

5.3.3 Limited Biometric or Hardware Integration

Current implementation does not include biometric attendance devices, RFID cards, or facial recognition systems. These features can enhance automation but require additional integration.

5.3.4 NLP Advisor Accuracy Depends on Dataset

Although the NLP advisor performs well, some answers may be incomplete if the required document or course material is not available in the dataset. Accuracy can improve with more training data.

5.3.5 Performance under Very High Load

While the system handles moderate load efficiently, extremely large numbers of users (1000+) may require stronger servers or cloud optimization.

5.3.6 Limited Mobile App Support

CampusGPT is optimized for web usage. A dedicated mobile app is not yet developed, which may limit accessibility for some students.

5.3.7 Manual Data Entry Required Initially

Admin must manually upload data such as course outlines, lecture files, departments, and user records during first-time setup. Automation improves afterward, but initialization still requires effort.

These limitations do not reduce the usefulness of CampusGPT but indicate areas that can be enhanced to make the system even more powerful and scalable.

5.4 Suggestions for Future Enhancements

Although CampusGPT provides a strong foundation for an AI-powered campus management system, there are several opportunities to extend and improve the platform in future versions. The following enhancements can make the system more powerful, user-friendly, and suitable for large-scale deployment.

5.4.1 Integration of Biometric and IoT Devices

Future versions can include biometric attendance systems, RFID cards, NFC tags, or facial recognition.

These integrations will automate attendance completely and make data more accurate.

5.4.2 Development of a Mobile Application

A dedicated Android and iOS mobile app can make the system more accessible for students, teachers, and staff. Push notifications and mobile-friendly dashboards can enhance user engagement.

5.4.3 Expanded AI Training with Real University Data

Training AI models using real, large-scale university datasets will further improve prediction accuracy, NLP responses, and quiz generation quality.

5.4.4 Voice-Based Virtual Assistant

Adding a voice-enabled AI assistant can allow users to interact using spoken language, making the system more inclusive and faster to use.

5.4.5 Advanced Analytics and Reporting Tools

Future versions can include deeper insights such as dropout prediction, teacher performance analytics, sentiment analysis of student queries, and automatic detection of struggling students.

5.4.6 Integration with Learning Management Systems (LMS)

CampusGPT can be connected with popular LMS platforms such as Moodle, Google Classroom, or Blackboard to create a unified learning and management environment.

5.4.7 Offline Mode for Low-Internet Areas

Adding limited offline functionality can help students and staff in areas with slow or unstable internet connections.

5.4.8 Multi-Language Support

Supporting Urdu and other regional languages will make the system more user-friendly for local universities in Pakistan.

5.4.9 Chat-Based Workflow Automation

More AI agents can be added to handle tasks like automatic timetable generation, fee reminders, attendance forecasting, and assignment scheduling.

5.4.10 Cloud Auto-Scaling

Deploying the system on cloud platforms with auto-scaling will ensure smooth performance even under very high user load.

ORIGINALITY REPORT

10%	9%	3%	3%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	ijsmr.in Internet Source	1%
2	betanet.net Internet Source	1%
3	Submitted to Universiti Teknologi MARA Student Paper	1%
4	www.rapidinnovation.io Internet Source	1%
5	eprints.utm.my Internet Source	1%
6	iprojectmaster.com Internet Source	<1%
7	zealjournals.com Internet Source	<1%
8	umatechnology.org Internet Source	<1%
9	www.srjis.com Internet Source	<1%

10	Submitted to Pathfinder Enterprises Student Paper	<1 %
11	Submitted to The British University in Dubai Student Paper	<1 %
12	d-nb.info Internet Source	<1 %
13	edubirdie.com Internet Source	<1 %
14	centermast.com Internet Source	<1 %
15	skillapp.co Internet Source	<1 %
16	www.internetreputation.com Internet Source	<1 %
17	globusedujournal.in Internet Source	<1 %
18	ir.unisa.ac.za Internet Source	<1 %
19	www.dochub.com Internet Source	<1 %
20	dr.ntu.edu.sg Internet Source	<1 %
21	fastercapital.com Internet Source	<1 %

22	www.diva-portal.org Internet Source	<1 %
23	Shi, Lehong. "Exploring the Emerging Roles of Artificial Intelligence in Teaching Practices and Its Relationships with Teachers", University of Georgia, 2023 Publication	<1 %

Exclude quotes	On	Exclude matches	< 3 words
Exclude bibliography	On		